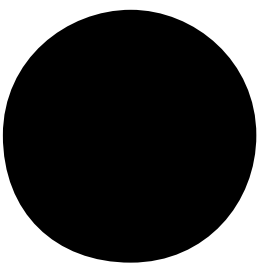
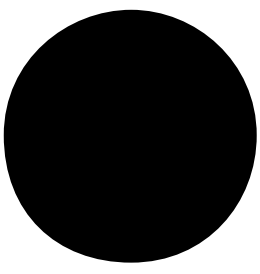
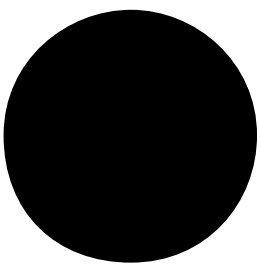
What is a neural network?

- A neural network is a way to parametrize a **function** of many variables in a computer in a **useful manner** in terms of many numbers: $W_{ij}^{(n)}, b_i^{(n)}$







x_i

$$x_i^{(1)} = \tanh(W_{ij}^{(1)} x_j + b_i^{(1)})$$

$$x_i^{(n)} = \tanh(W_{ij}^{(n)} x_j^{(n-1)} + b_i^{(n)})$$

$$f(x)$$

“Sailboat”

“It was the best of times…”

“…it was the worst of times.”

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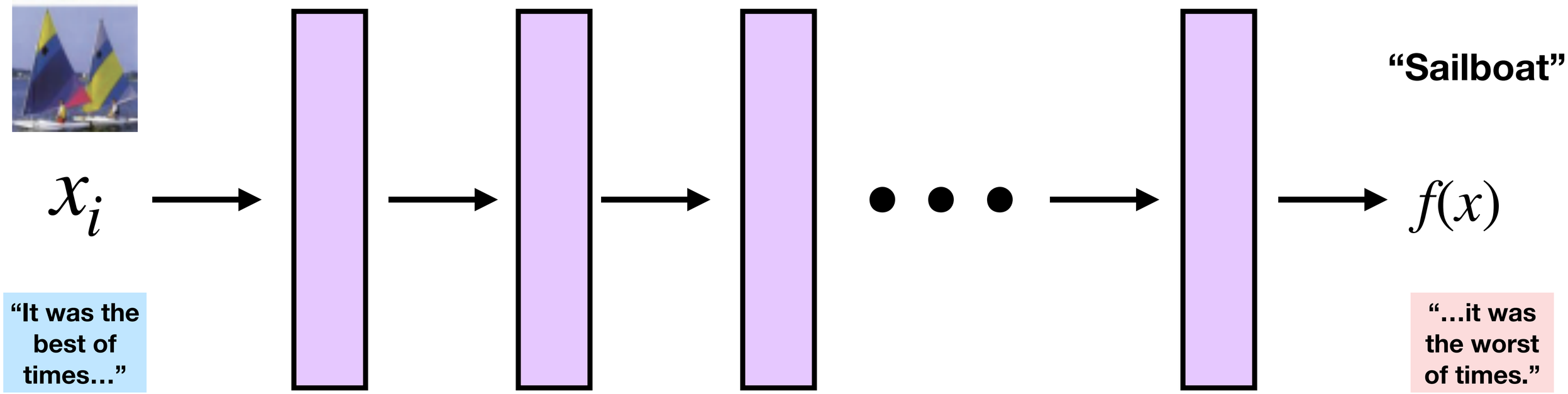
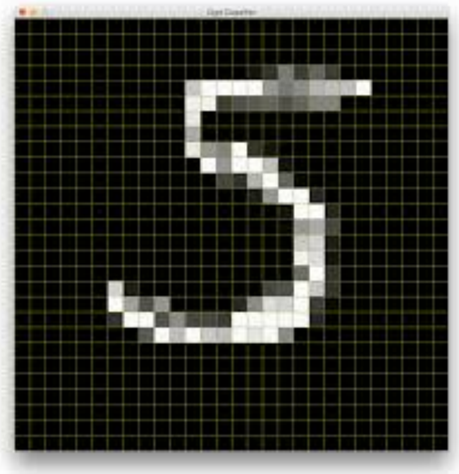


$$x_i^{(1)} = \tanh(W_{ij}^{(1)} x_j + b_i^{(1)})$$

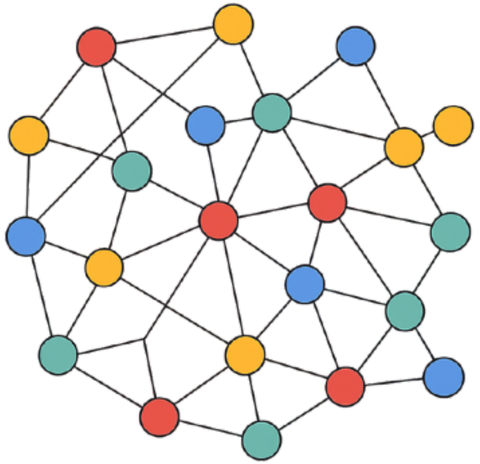
$$x_i^{(n)} = \tanh(W_{ij}^{(n)} x_j^{(n-1)} + b_i^{(n)})$$

How to organize data?

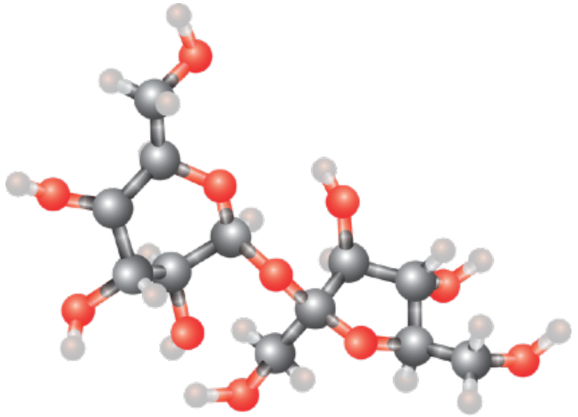
A universal way to think about different modalities of data is in terms of their **symmetries, e.g.**



Images: translations of the pixel grid.



Graph: permutations of the nodes and edges.



Molecules: translations and rotations of the particle coordinates.

We should use these symmetries to organize our neural networks.